Wiem Ferchichi

Embedded Systems, AI and Robotics | Final-Year Internship

Tunis, Tunisia | wiem.ferchichi@enstab.ucar.tn | +216 99 952 627 | LinkedIn | GitHub

PROFILE Advanced Technologies engineering student specializing in advanced electronics and nanotechnology, with two internship experiences in applied AI and projects combining embedded systems, IoT, robotics and simulation. Seeking a final-year internship in advanced technologies, intelligent systems or embedded systems.

EDUCATION

ENSTAB – National School of Advanced Sciences and Technologies of Borj Cedria Tunis, 2024–Present
Engineering Degree in Advanced Technologies

Faculty of Sciences of Monastir Monastir, 2022–2024
Preparatory Program in Mathematics and Physics (MP) – Diploma obtained

PROFESSIONAL EXPERIENCE

OpenCloud – Software Development Company Tunis, June–August 2026
Local AI Technical Assistant (LLM + RAG)

- Developed a local document question-answering system using Python, Flask, Ollama and ChromaDB.
- Used embeddings and vector search to retrieve relevant passages for a local language model, with a conversational interface.

LARINA – Research Laboratory in Smart Grids and Nanotechnology Tunis, June–August 2025
Reinforcement Learning for Urban Energy Management

- Conducted reinforcement learning experiments for building energy management using CityLearn.
- Evaluated control behavior in simulation.

PROJECTS

ENSTAB – Second-Year Engineering Project Jan.–May 2026
4-DOF Robotic Arm and Digital Twin

- Controlled the motion of a 4-DOF robotic arm, with sensor feedback and serial communication between the controller and actuators.
- Developed a URDF model and PyBullet simulation for virtual validation and trajectory optimization.

ENSTAB – First-Year Engineering Project Jan.–May 2025
Smart Home Security System (IoT)

- Designed an IoT architecture on Raspberry Pi, including sensors, edge processing and monitoring, using MQTT for sensor-to-actuator communication.
- Integrated a CNN for real-time intrusion detection.

Personal Projects 2025–2026
Edge ML, Mobile Robotics and FPGA

- Classified sensor data using Edge Impulse and deployed models on ESP32.
- Developed a four-wheeled mobile robot with ultrasonic obstacle detection.
- Designed, simulated and implemented VHDL circuits on an FPGA board using Vivado.

TECHNICAL SKILLS

AI and Edge AI: Machine Learning, CNNs, computer vision, reinforcement learning, local LLMs, RAG, ChromaDB, Edge Impulse
Embedded Systems and IoT: ESP32, STM32, Raspberry Pi, Arduino, C/C++, Python (intermediate), MQTT, UART, I^2C, LoRa
Robotics and Simulation: URDF, PyBullet, sensor integration, PID control
Design and Tools: VHDL, FPGA, Xilinx Vivado, Linux, Flask, Ollama, CityLearn, MATLAB/Simulink

INTERPERSONAL SKILLS

Leadership and Teamwork: coordinated project teams within a junior enterprise, collaborating with technical professionals and clients
Communication: technical training, hackathon support, client relations (prospecting and negotiation)
Problem-Solving, Autonomy and Attention to Detail: contributed to projects from design and simulation through hardware validation
Adaptability: learning new technologies in AI, embedded systems, FPGA and robotics; Agile/Scrum methodologies

EXTRACURRICULAR ACTIVITIES

ETC Junior Enterprise – Project Manager 2025–2026
Project Coordination and Team Collaboration

- Led 7 internal projects in embedded systems and energy with a team of 5+ members; contributed to client projects involving prospecting, planning and delivery.
- Organized technical and professional training sessions.

Electronix Club – Active Member 2025–2026
Trainer and Technical Support

- Provided technical support to members during hackathons, trained peers and served on internal competition juries.

CERTIFICATION AND LANGUAGES

ISO 9001:2015 Quality Management (April 2024) | Arabic (Native) | French (Fluent) | English (Fluent)